Jared Nussbaum

Senior DevOps Engineer

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PROFILE SUMMARY

Senior DevOps Engineer with 5+ years of experience architecting multi-account AWS data platforms, managing research Linux HPC infrastructure, and building systems automation. An outspoken voice for sound technical direction, focused on communicating architectural trade-offs, asking critical questions, and aligning systems with business goals. Hands-on, high-throughput engineer focused on selecting the right technology to build and ship robust tools, platforms, and software systems.

WORK EXPERIENCE

Senior DevOps Engineer | **Federal Reserve Bank of Boston** June 2021 - Present • Boston, MA

- **AWS Data Platform:** Architected and built an AWS Data Platform using Terraform and Ansible, managing Apache Airflow data pipelines, DataHub cataloging, automated CI/CD releases, and dynamic Okta-governed IAM access controls across all 12 Federal Reserve research districts.
- **AI FinOps & Cloud Observability:** Engineered an AWS Bedrock cost observability dashboard utilizing AWS Cost and Usage Reports (CUR 2.0) and Streamlit deployed on Amazon ECS, delivering daily telemetry on per-user, per-model, and per-token-type LLM consumption and spend.
- **Authentication Tooling:** Developed an enterprise Okta authentication client utilizing AI agent workflows to enable seamless, secure AWS role assumption via ADFS, Entra ID, and Okta Authentication supporting both Classic and OIE identity architectures.
- **Cross-Language Data CLI:** Engineered a cross-language Athena data retrieval CLI, packaging custom Python, R, Julia, Stata, and MATLAB FFI wrappers for automated AWS credential negotiation and native DataFrame ingestion.
- **HPC Workload Modernization & Capacity:** Maintain, modernize, and stabilize the research Linux HPC system, leading the workload scheduler migration from Sun Grid Engine (SGE) to Slurm and guiding hardware capacity and purchasing decisions, while implementing cgroups, expanding cluster accounting, and ensuring zero disruption to ongoing research workflows.
- **Infrastructure Automation & CI/CD:** Automated VMware vSphere virtual machine and bare-metal server provisioning using Packer, Terraform, Ansible, PXE Boot, and Kickstart, while orchestrating CI/CD pipelines for automated service deployments.
- **Containers & Observability:** Containerized in-house internal tools and COTS software using Docker, while managing full-stack Prometheus and Alertmanager observability infrastructure with Jira integrations to proactively track cluster health and uptime.

Engineering Intern | **CPX Security** Summer 2019, 2020 • Boston, MA

- Engineered scalable, Docker-based IoT device simulations to test integration and security across Azure and Pelion cloud platforms.

IT DevOps & Solutions Development Intern | **Raytheon IDS** Summer 2017, 2018 • Tewksbury, MA

- Standardized containerized Machine Learning environments using Docker and Anaconda, and built CI/CD automation pipelines to streamline build, test, and deployment lifecycles.
- Led an IT asset lease tracking optimization project driving an estimated $100,000+ in cost savings, and deployed enterprise binary artifact storage (Artifactory) on Red Hat Enterprise Linux (RHEL).

PROJECTS

Adaptive Space (adaptivespace.ai) 2026

- 100% client-side, local-first progressive web app for runners. Combines deterministic exercise physiology guardrails with an agentic AI coach that reconciles schedules against activity and health telemetry, persisting data and reports in local IndexedDB.

Adaptive Pace Calculator (calc.adaptivespace.ai) 2026

- Continuous mathematical pacing and altitude adjustment engine implementing Daniels/Gilbert oxygen demand quadratic models, Bassett altitude polynomial scaling, and a numerical bisection sea-level solver with 23 automated tests.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS, Amazon ECS, AWS Bedrock, Terraform, Ansible, Packer, Docker, Linux HPC, Slurm, VMware vSphere, Prometheus, Alertmanager, PXE Boot, Kickstart, Apache Airflow, DataHub, NVIDIA CUDA, IAM, Okta
Languages & Automation: Python, Streamlit, Linux Bash / Shell, PowerShell, SQL, Athena, PostgreSQL, C / C++, Git, GitLab CI, Conda, Pixi

EDUCATION

University of Massachusetts Amherst Amherst, MA
B.S. in Computer Science & Political Science Sept 2016 - May 2020
GPA: 3.92 / 4.0 • Magna Cum Laude • Dean's List Honors (All Semesters)